



Measuring the radius of circular electron beams to determine the charge-to-mass ratio of the electron

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Electrons moving through a uniform magnetic field in the absence of an electric field experience a force resulting in circular motion. The radius of the circular motion due to the energy put into the electrons from the electron gun, and the strength of the applied magnetic field are related to the ratio of the charge of an electron to its mass. The squared radius of the curvature of the electron beam is found to be linearly related to the ratio of voltage and squared magnetic field strength by a constant term related to the charge to mass ratio of the electron. The measured electron charge to mass ratio is determined to be $(1.8 \pm 0.2) \times 10^{11}$ C/kg.

I. INTRODUCTION

Not long ago, atoms were considered indivisible. The discovery of the charge-to-mass ratio of the electron played a pivotal role in our understanding of subatomic physics. This groundbreaking result contributed to the identification and characterization of electrons as distinct entities within atoms, laying the foundation for the development of atomic and particle physics [1]. Additionally, the precise determination of the charge-to-mass ratio also led to developments in cathode ray tube technologies, opening new avenues for research and technological applications [2].

This report seeks to measure the charge-to-mass ratio of the electron. Electrons accelerated by an electron gun will undergo a force due to a uniform magnetic field perpendicular to their motion, resulting in circular motion. Precise measurements of the magnetic field, voltage applied to the electron gun, and radius of the resultant circular motion will yield the electron charge to mass ratio.

A fundamental physics principle governing circular motion in the experiment is the Lorentz force, describing the movement of charged particles within electric and magnetic fields [3]. In this specific experiment, solely a magnetic field is present, with no applied electric field. The resulting force is perpendicular to both the magnetic field lines and the particle's motion. As the particle accelerates inward, the centripetal force will increase until equilibrium is achieved, resulting in sustained circular motion.

II. KEY FINDINGS

A. Mathematical Background

In order to discover the charge-to-mass ratio of the electron, it is important to understand how fields affect particles, particularly magnetic fields. Recalling the Lorentz force, an electron with mass m_e , charge q_e , and velocity v perpendicular to a uniform magnetic field of strength B in the absence of an electric field will experience a force of magnitude

$$F = q_e v B. \quad (1)$$

in a direction perpendicular to the plane formed by the velocity and magnetic field directions. Due to this perpendicular force, the electron will undergo circular motion

$$\frac{m_e v^2}{r} = q_e v B, \quad (2)$$

$$v^2 = \left(\frac{q_e}{m_e} \right)^2 B^2 r^2, \quad (3)$$

where r is the radius of the circle formed by the electrons. During motion, the electron's kinetic energy is equal to the energy gained from the electron gun with voltage V_{gun} that initially accelerated the electrons

$$\frac{1}{2} m_e v^2 = q_e V_{\text{gun}}. \quad (4)$$

Combining eq. (3) and eq. (4) yields the following relation

$$r^2 = \left(2 \frac{m_e}{q_e} \right) \frac{V_{\text{gun}}}{B^2}, \quad (5)$$

which is a linear relationship between the radius squared and the ratio between voltage and the squared magnetic field strength by a factor related to the charge-to-mass ratio of the electron.

B. Methods

The experiment was conducted using the Uchiga TG-13 apparatus which includes an electron gun to produce a beam of electrons, two Helmholtz coils to produce a controlled magnetic field, and an evacuated tube with a trace amount of helium that will produce a glow when electrons move through the gas.

The electron gun is made up of two electrodes connected to a voltage source of magnitude V_{gun} . The cathode is heated up which allows for electrons to escape its surface more easily. The positively charged anode has a hole in the center and accelerates the free electrons through its center which creates a focused beam of electrons.

The two Helmholtz coils are loops of wire placed around the experiment apparatus, sandwiching the vacuum tube. When the coils are wired in series so that the current flowing through both coils flows in the same direction, a region with a nearly uniform magnetic field is produced in the center [4]. Knowledge of the number of turns in the coils, radius of each coil, and distance between the coils provides a conversion between current flowing through the coils and the magnetic field strength produced at the center of the vacuum tube.

The radius of the circular motion of the electrons was determined by a ruler positioned behind the vacuum tube. The ruler is reflective and allows the user to line up their eye with the reflection of their eye to account for parallax. Measuring the position of each edge of the circle formed by the electrons allows the user to measure the diameter of the circle and thus the radius.

The data collected includes measurements of varying voltages applied to the electron gun, varying strengths of the magnetic field, and the resultant radius of the glowing circle formed by the electron beam. Least squares regression between the squared radius and the ratio of voltage to squared magnetic field as described in eq. (5) will yield a slope χ that is related to the charge-to-mass ratio in the following manner

$$\frac{q_e}{m_e} = \frac{2}{\chi}. \quad (6)$$

Linear regression is preferred over repeated calculations of the charge-to-mass ratio because individual ratio calculations, being based on mean values, are more susceptible to systematic errors. They fail to account for the overall data distribution and variations in both independent and dependent variables, making them sensitive to outliers and potentially leading to skewed results. Least squares regression minimizes such errors by fitting a line to the entire dataset.

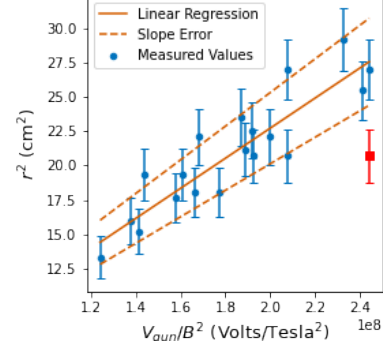


FIG. 1. The measured radii squared is plotted against the measured ratio of voltage to squared magnetic field strength. The solid trend-line is a least squares regression fit with dashed lines representing the error in the slope. An outlier (square point) is highlighted in red and not included in regression.

C. Results

The plot of measured values for radius squared versus the ratio between electron gun voltage and squared magnetic field strength is shown in fig. 1. The calculated value of the charge-to-mass ratio for the electron is

$$\frac{q_e}{m_e} = (1.8 \pm 0.2) \times 10^{11} \frac{\text{C}}{\text{kg}}. \quad (7)$$

The true charge-to-mass ratio of the electron is within the range of 1.6×10^{11} to $2 \times 10^{11} \frac{\text{C}}{\text{kg}}$. In the limit of infinite magnetic field the radius of the circular electron beam should tend towards zero, which is confirmed by our computation that yields

$$\lim_{B \rightarrow \infty} r^2 = 1 \pm 2 \text{ cm}^2, \quad (8)$$

which includes zero radius due to the standard error in the calculation. This is representative of the offset in our measured values of r^2 .

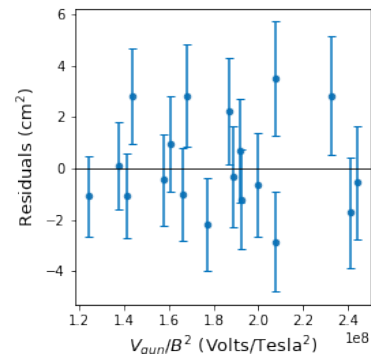


FIG. 2. Plot of the residuals, the differences between measured and predicted values, of the square of the radii.

It is important to note that there was an outlier found to be present within the data that was removed from the linear regression computation. This point was determined to be more than three standard deviations from the trend line and was therefore discarded.

The plot of the residuals [fig. 2], which depicts the measured value minus the predicted value, shows a randomly distributed pattern indicating a linear relationship. Additionally, the correlation coefficient R between the measured and predicted data shows a strong linear correlation:

$$R^2 = 0.90. \quad (9)$$

The measured value of the charge-to-mass ratio has a large degree of error likely due to the measurement techniques. Data was collected by five separate people throughout the course of the collection process and synthesized into one data set. Due to differences in measurement style and each person's varying methods of dealing with parallax, the error in the charge-to-mass ratio is large in comparison to the measured value (roughly $\pm 11.9\%$).

III. CONCLUSION

The balance between the Lorentz force and the centripetal force leads to circular motion of the electron beam. The energy provided by the electron gun is transferred into kinetic energy and a linear relationship was found between the squared radius of the curvature of the electron beam and the ratio of voltage to squared magnetic field strength by a constant term related to the charge to mass ratio of the electron. The charge-to-mass ratio of the electron was measured to be $(1.8 \pm 0.2) \times 10^{11} \frac{\text{C}}{\text{kg}}$, as determined by linear least squares regression.

To produce a more precise result, averaging two independent measurements of the radius of the electron motion would significantly decrease the error in radius measurements. Also, consistency in measurements by the same individuals helps mitigate bias in parallax, as measurement methods vary across participants. Furthermore, an investigation into incorporating relativistic kinetic energy into the model may provide increased accuracy due to the substantial acceleration induced by high voltages.

Appendix A: Uncertainty Analysis

The uncertainties in each measurement of length L , voltage V , and current I are shown below

$$\delta_L = 0.003 \text{ m} \quad (A1)$$

$$\delta_V = 1 \text{ V} \quad (A2)$$

$$\delta_I = 0.02 \text{ A.} \quad (A3)$$

The radius of the electron circle was calculated by averaging the two length measurements (one on either side of the circle) to account for the center of the electron circle and the zero point on the ruler being offset. The uncertainty in the radius measurement is shown below

$$\delta_r = \frac{\sqrt{2}}{2} \delta_L, \quad (A4)$$

$$= 0.002 \text{ m} \quad (A5)$$

Because we are plotting the radius squared, the uncertainty in each radius squared calculation is

$$\delta_{r^2} = 2r\delta_r, \quad (A6)$$

The magnetic field generated by the pair of Helmholtz coils was calculated by the conversion factor given by the experiment apparatus

$$B(I) = 7.80 \times 10^{-4} I. \quad (A7)$$

Therefore the uncertainty in the magnetic field is

$$\delta_B = 7.80 \times 10^{-4} \delta_I. \quad (A8)$$

The uncertainty in the voltage divided by the squared magnetic field strength is given by

$$\delta_{V/B^2} = \frac{1}{B^2} \sqrt{\delta_V^2 + 4 \frac{V^2}{B^2} \delta_B^2}. \quad (A9)$$

which can be derived from propagation of uncertainty (adding the partial derivatives multiplied by their uncertainties in quadrature).

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